

SOFTWARE ENGINEERING LAB-1

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Section:- B

Task:- Requirements Engineering & UML Use-Case Modelling

Course:- Software Engineering (UE24CS341A)

Problem Statement:-

Patient Health Record Consent Management System

Domain:-

Healthcare & Telemedicine

Actors:-

- Patient
- Clinic Administrator

Project Overview:-

The project is a Patient Health Record Consent Management System in the healthcare and telemedicine domain. The system focuses on patient-controlled, granular and time-bound access to health records.

Lab 1 Deliverables:-

The Lab 1 contains:-

1. Requirements Table

- Functional Requirements :- FR-001 to FR-005
- Non- Functional Requirements:- NFR-001 to NFR-002

2. UML Use-Case Diagram

- Represents the interactions between the Patient, Clinic Administrator, and the patient health record consent Management System.

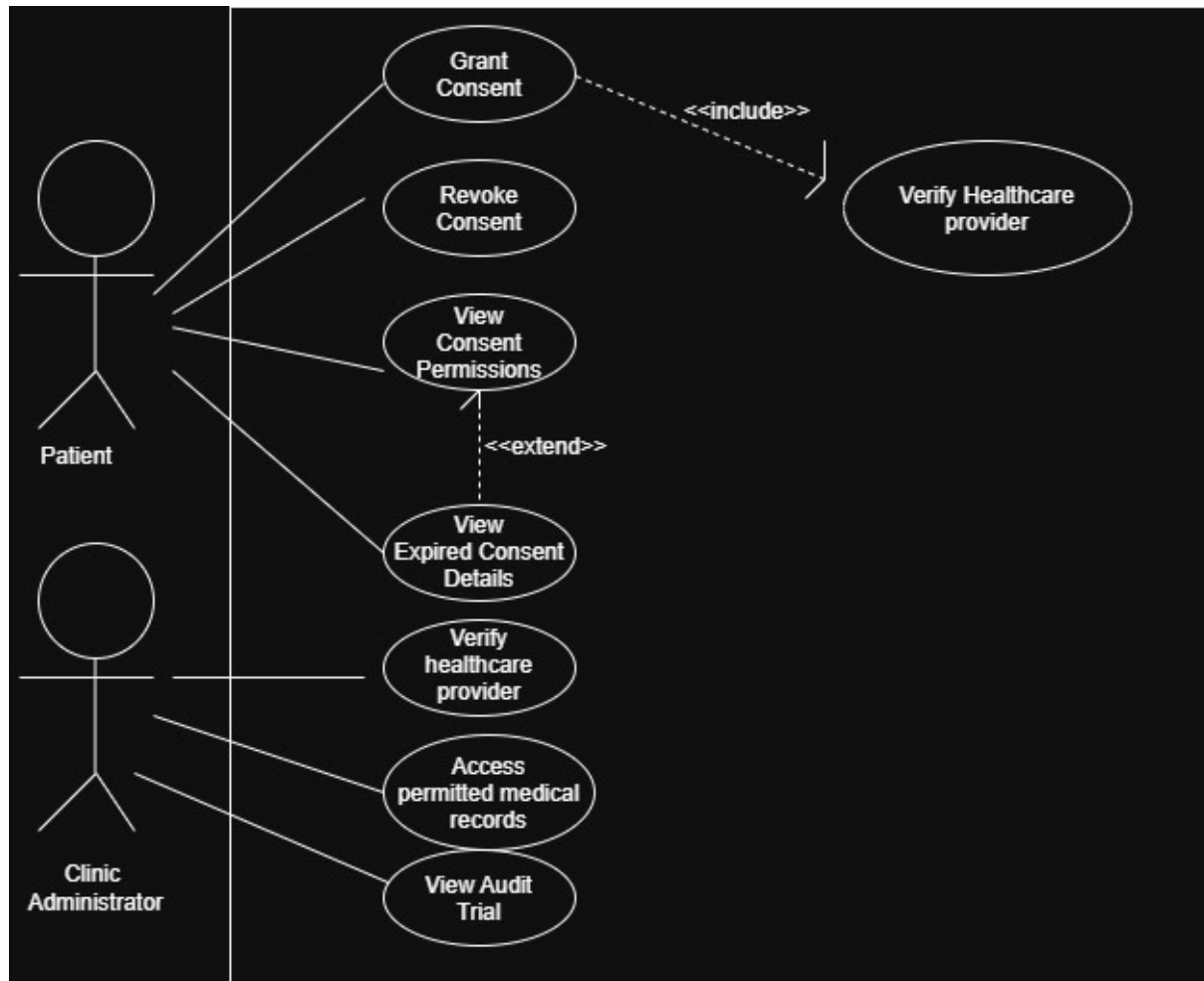
3. Use-Case Flow Specification

- Core use case:- Grant Time-Bound Consent
- Includes preconditions , postconditions, main case scenario, and alternate flow

1] Requirements Table:-

Requirements Table					
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow patients to grant time-bounded (e.g., 24-hour) access permissions for specific diagnostic records to verified clinic doctors.	High	Pass: Clinic access is automatically revoked upon the specified expiry timestamp. Fail: Clinic accesses records after consent expiration.	Allows patients to control access to specific diagnostic records for a defined period.
FR-002	Functional	The system shall allow patients to revoke an active consent before its scheduled expiry time.	High	Pass: The selected consent becomes inactive immediately after revocation and the associated records are no longer accessible. Fail: The healthcare provider can still access the records after revocation.	Gives patients control to withdraw previously granted access to their medical history.
FR-003	Functional	The system shall allow patients to view the consent permissions they have granted to clinics, diagnostic labs, and consulting doctors.	High	Pass: The patient can view the healthcare provider, permitted records, consent status, and expiry time. Fail: The system displays incomplete or incorrect consent information.	Enables patients to monitor and manage who has access to their medical history.
FR-004	Functional	The system shall allow authorized healthcare providers to access only the medical records covered by an active patient consent.	High	Pass: The provider can access only the records explicitly permitted by the patient while the consent is active. Fail: The provider can access records outside the granted permission or after consent expiry.	Enforces granular, patient-controlled access to sensitive health records.
FR-005	Functional	The system shall allow the Clinic Administrator to verify and manage authorized healthcare providers before they are permitted to access patient records.	Medium	Pass: The Clinic Administrator can verify provider details and only verified providers can receive patient-authorized access. Fail: An unverified provider is granted access to patient records.	Helps ensure that patient records are accessible only to legitimate healthcare providers.
NFR-001	Non-Functional	All consent grant, revoke, and record-access events shall be permanently written to an append-only audit trail.	High	Pass: Benchmarking and audit tests confirm that consent and access events are recorded and cannot be modified or deleted through normal system operations. Fail: An event is missing or can be altered.	Provides accountability, traceability, and security for access to sensitive health records.
NFR-002	Non-Functional	The system shall maintain secure and reliable consent enforcement while meeting the required response-time and performance standards under normal and peak usage.	High	Pass: Performance testing confirms that consent checks and access-control operations meet the defined response-time and security standards under simulated peak load. Fail: The system exceeds the defined performance or security limits.	Ensures that patient data remains protected without making the consent management system unreliable or unusably slow.

2] UML Use-Case Diagram:-The UML use-case diagram should model interactions between the Patient, Clinic Administrator, and the Patient Health Record Consent Management System. Both actors should be placed on the left side of the system boundary



3] Core Use-Case Flow- Grant Time-Bound Consent

Use-Case Flow Specification

Use Case: Grant Time-Bound Consent

Primary Actor: Patient

Supporting Actor: Clinic Administrator

Preconditions

1. The patient is registered and authenticated in the system.
2. The patient has access to their health records.
3. The healthcare provider is registered and can be verified by the system.

Postconditions

1. A consent permission is created for the selected healthcare provider.
2. The selected diagnostic records are accessible only for the specified consent period.
3. The consent automatically becomes inactive when the expiry time is reached.
4. The consent grant event is recorded in the audit trail.

Main Success Scenario

1. The patient logs into the Patient Health Record Consent Management System.
2. The patient selects the diagnostic records they want to share.
3. The patient selects the healthcare provider who should receive access.
4. The patient specifies the duration or expiry time for the consent.
5. The system verifies the selected healthcare provider.
6. The system displays the selected records, provider, and consent duration for confirmation.
7. The patient confirms the consent.
8. The system creates the time-bound consent permission.
9. The system records the consent grant event in the audit trail.
10. The system confirms that the consent has been successfully granted.

Alternate Flow

A1. Healthcare provider cannot be verified

1. At Step 5, if the selected healthcare provider cannot be verified, the system rejects the consent request.
2. The system informs the patient that the provider could not be verified.
3. No consent permission is created.
4. The patient may select another verified healthcare provider and restart the consent process.

5] Conclusion:-

The Patient Health Record Consent Management System enables patients to control access to medical records through granular and time-bound consent. The requirements define the expected functionality and quality attributes, while the UML use-case model represents interactions between the identified actors and the system.